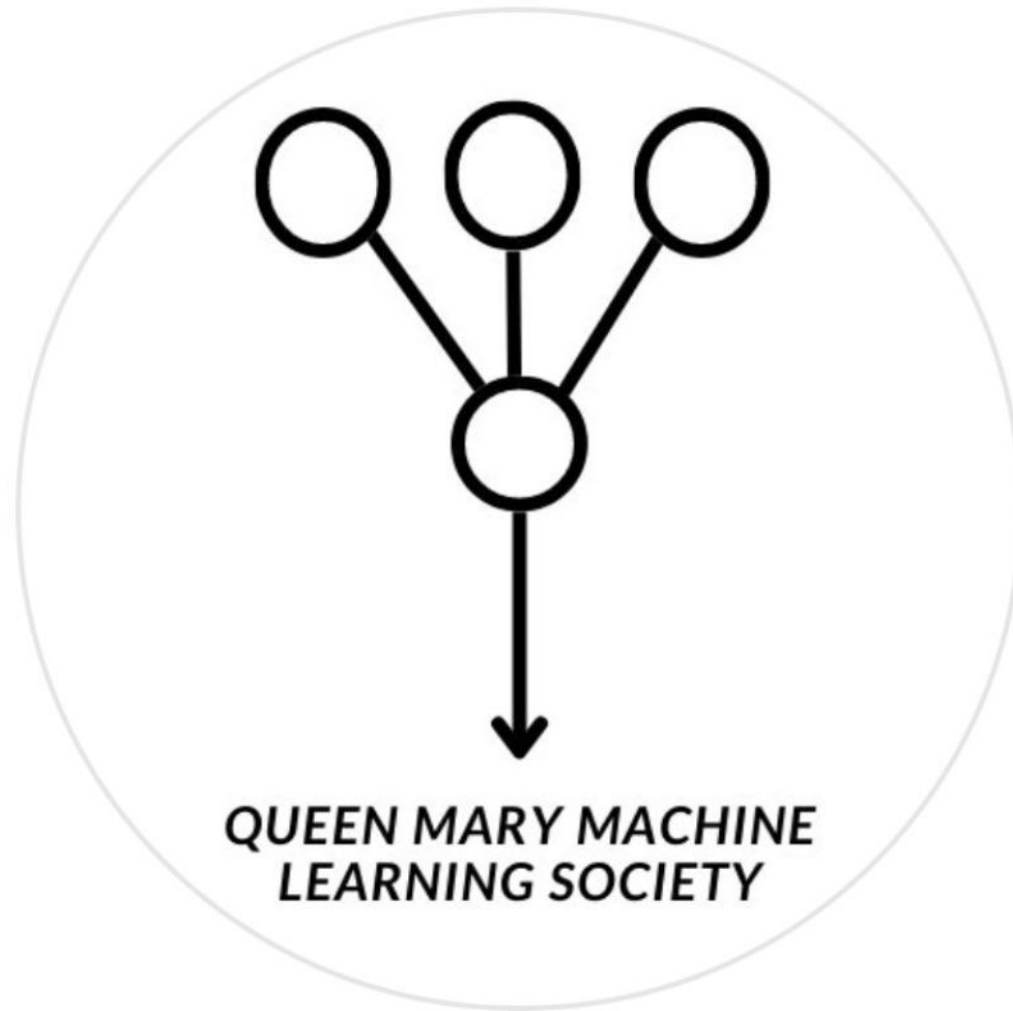


Kaggle #05

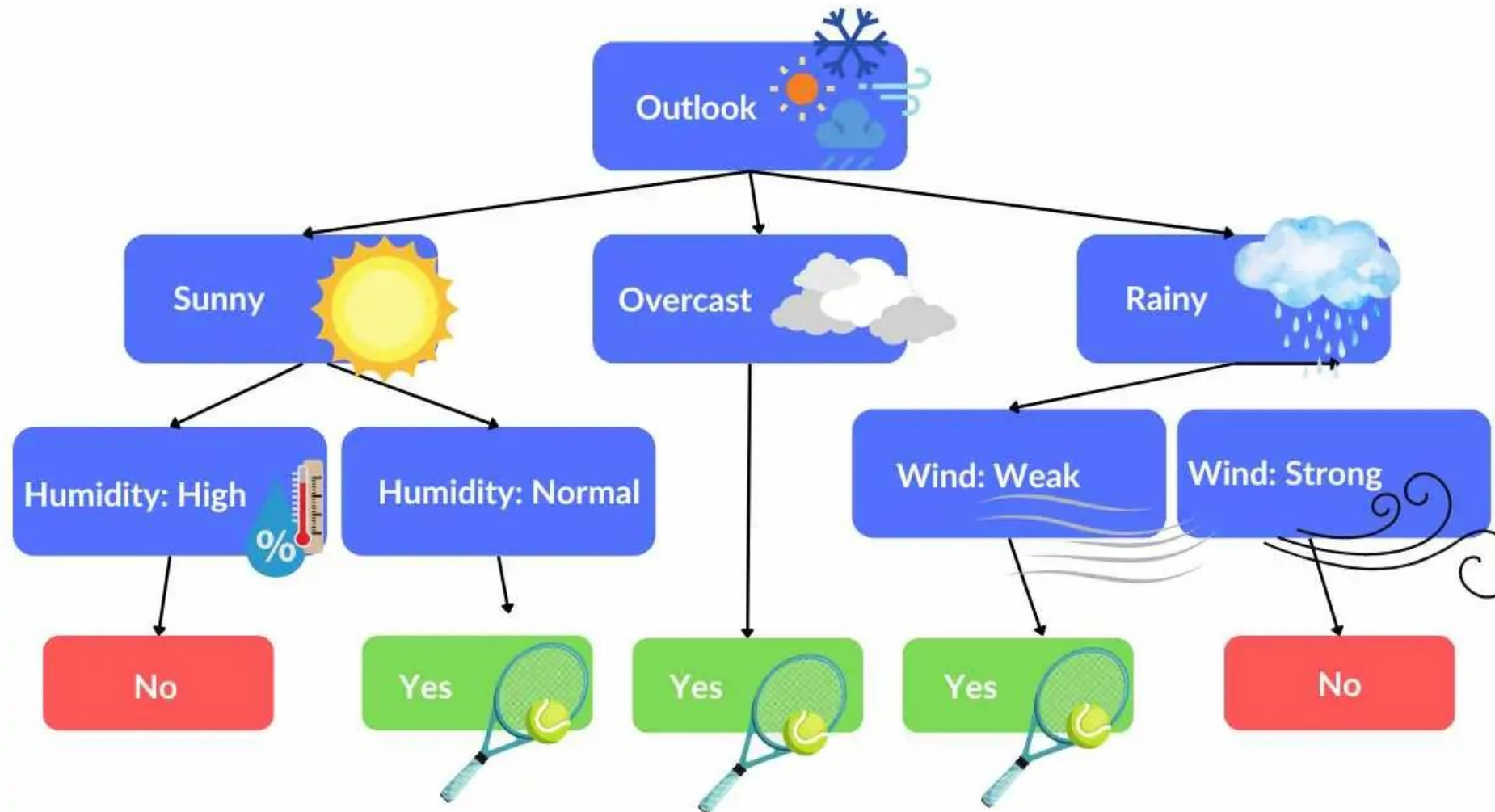


Decision Trees

- A supervised learning model that uses a series of if/else rules to make predictions
- Splits data into branches based on the feature that best separates the target
- Each internal node = a question (e.g., “Income > 30k?”)
- Each leaf node = a prediction
- Works for both classification and regression
- Intuitive, human-readable structure similar to a flowchart

Decision Trees

Are we playing tennis



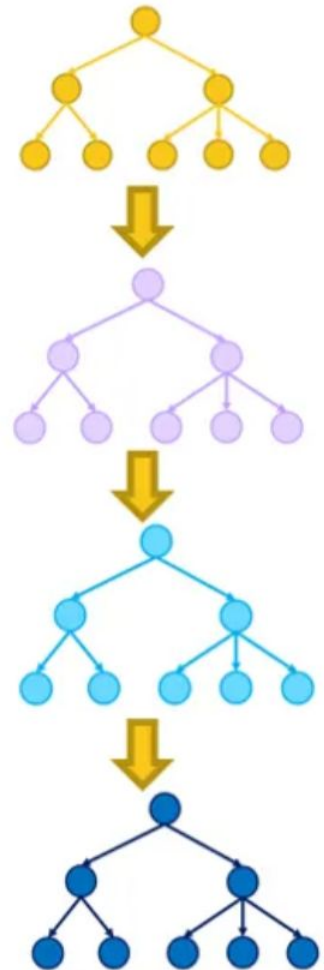
Pros and Cons of Decision Trees

Advantages	Disadvantages
Easy to understand and explain	Can overfit easily (if too many levels to tree)
No scaling or normalisation needed	High variance (small changes, big impact)
Can handle both numerical and categorical features	As single model, is has weak predictive power
Captures non-linear relationships	Hard to catch onto subtle trends without ensembles

One tree may be weak, but a forest of trees can be powerful

What if we use many trees?

- A single decision tree is unstable and often overfits
- But many weak trees can be combined to form a strong, stable model
- Multiple trees working together can:
 - Reduce errors
 - Capture more patterns
 - Improve generalisation
- This idea leads to Ensemble Learning...



Ensemble Learning (Brief Intro)

- **Ensemble Learning** - combining multiple models to produce a stronger predictor. There are three main types:
- **Bagging**
 - Parallel, independent training of different models on different subsets of data
 - Final prediction = **Average** of all models for regression, or **Majority vote** for Classification
- **Boosting**
 - Sequential training. Each new model fixes errors of the previous
 - Final prediction = **Sum** (weighted) of models
- **Stacking / Blending**
 - Combine predictions from different model types using a meta-model (Quite advanced!)

What is Boosting?

Boosting - building many weak models sequentially, where each one learns from the errors of the previous.

How it works (simplified):

1. Train a simple model (usually a small decision tree)
2. Measure its errors (residuals or incorrectly classified samples)
3. Train the next model to correct those errors
4. Repeat → each model focuses more on the “hard” cases
5. Final prediction = weighted sum of all models

$$\text{Prediction}(x) = f_1(x) + f_2(x) + \dots + f_M(x)$$

Pros and Cons of Boosting

Pros	Cons
Reduces bias	Can overfit if not tuned
Turns weak learners into a strong model	More complex than bagging
Learns from mistakes → improves each round	Slower training than Random Forest
Handles non-linear patterns well	Less interpretable than a single tree
Excellent accuracy on tabular data	Sensitive to noisy data / outliers
Doesn't plateau early - keeps improving	More hyperparameters to tune

Why use Boosting in Kaggle?

- Excels on tabular data
- Handles mixed feature types (categorical + numeric)
- Captures non-linear relationships
- Automatically models feature interactions
- Deals well with missing values
- Produces well-calibrated probabilities
- Is fast, scalable, and highly optimised (LGBM, XGBoost, CatBoost)
- Continues improving with more iterations (doesn't plateau early)

Popular Boosting models

The Big 3:

- XGBoost
- LightGBM (LGBM)
- CatBoost

Why these dominate Kaggle:

- Extremely powerful out of the box
- Handle missing values, categorical features, and non-linear relationships
- Strong generalisation on tabular data
- Built-in regularisation to reduce overfitting
- Proven track record in Kaggle competitions

TLDR

- **Decision Trees** are simple but weak and unstable
- **Ensemble Learning** combines many trees to build stronger models
- **Bagging** stabilises models (Random Forest)
- **Boosting** improves models by fixing errors sequentially
- Boosting achieves **higher accuracy** on tabular data
- The “Big 3” — **XGBoost, LightGBM, CatBoost** — dominate Kaggle
- These models handle **nonlinearity, feature interactions, missing values & categorical data** extremely well

NEW Playground Series



KAGGLE · PLAYGROUND PREDICTION COMPETITION · A MONTH TO GO

Join Competition



Predicting Loan Payback

Playground Series - Season 5, Episode 11



Overview Data Code Models Discussion Leaderboard Rules

Overview

Welcome to the 2025 Kaggle Playground Series! We plan to continue in the spirit of previous playgrounds, providing interesting and approachable datasets for our community to practice their machine learning skills, and anticipate a competition each month.

Your Goal: Predict the probability that a borrower will pay back their loan.

Start

5 days ago



Close

25 days to go

Competition Host

Kaggle



Prizes & Awards

Swag

Does not award Points or Medals

Participation

2,927 Entrants

827 Participants

811 Teams

4,193 Submissions

Tags

Feedback form

